SE(3)-WAM: Role-Aware 4D World–Action Modeling for Dynamic Manipulation

Anonymous CVPR submission

Figure placeholder: Fig. 1 teaser — a role-aware 4D view of a moving scene (real robot)

Layout: full width, two parts. **(a) Left, ∼62%:** one real-robot frame of the *linear-motion grasp* at the instant a chunk is issued. Overlays in the paper's semantic colors: blue = the target's past SE(3) poses from the causal tracker (small triads at $-350 \ldots 0$ ms, fading with age); orange = the target's SE(3) forecast at the execution times of the chunk, taken from the training-time heads evaluated offline; light orange arrows = sparse 3D flow on the 14×14 grid over the whole scene; purple = the commanded gripper path meeting the target at τ^{a}. A gray ghost marks the pose observed at capture, with a dashed red arrow showing where a reactive policy would close. A thin timeline strip under the image: capture $\tau_t \rightarrow$ latency $\ell_t \rightarrow$ execution τ^{a}. **(b) Right, ∼38%:** top: "Image-space WAM: future frames or latents at uniform fidelity" (filmstrip icon); middle: "SE(3)-WAM: SE(3) for the target, sparse 3D flow for the whole scene, training only" (triads + sparse arrows); bottom **(required)**: SR vs. added latency (0–200 ms) for SE(3)-WAM and the strongest baseline (data: the latency sweep of Appendix D.2). **Rules:** real photo kept unretouched; overlays drawn from logged tracker/forecast/command data of that trial; caption states the trial's outcome.

Figure 1. **A role-aware 4D view of a moving scene.** Between image capture and the first command, a part on the conveyor moves farther than the grasp tolerates. SE(3)-WAM encodes the target's metric motion (blue) and is trained to forecast the target's SE(3) pose precisely and the whole scene coarsely as a sparse 3D flow (orange), so that its commands meet the target when they execute (purple). Image-space world–action models forecast appearance at uniform fidelity instead. Right: success against added latency on Dynamic-14.

Abstract

Manipulating moving objects requires actions that fit the scene when they execute, not when the image was captured, yet every millisecond spent inferring the scene's motion lets the target move further. World–action models forecast future images or latents at uniform fidelity, spending time on appearance while leaving the target's metric pose implicit; target-centric predictors are fast but ignore the surroundings that constrain or drive the motion. We observe that the parts of a dynamic scene need different precision: the target's pose decides contact to within millimeters, the bodies around it matter only through the clearance they leave and the motion they impart to the target, and the remaining background does not matter. We present SE(3)-WAM, a world–action model built on this role-aware 4D structure. It encodes the target's motion as a metric SE(3) pose history and is trained to forecast the target's future SE(3) trajectory precisely and the whole scene coarsely as a sparse 3D flow. The forecasts serve only as training supervision and are skipped at deployment, so inference needs neither past frames nor a generated future and stays fast enough for real-time control. On Dynamic-14, a simulated benchmark in which the scene keeps moving during inference, SE(3)-WAM achieves [XX.X]% success, [XX.X] points above the strongest baseline, and succeeds in [XX]% of trials on six real-robot tasks with diverse target motion.

1. Introduction

Learning-based manipulation has advanced rapidly, from diffusion policies to vision–language–action (VLA) and world–action models (WAMs) [5, 9, 20, 31, 44, 50, 65], yet most of this progress assumes that objects stay still unless the robot moves them. In factories, kitchens, and human–robot handovers, objects keep moving while the robot perceives, computes, and acts. This poses two coupled difficulties. A single image does not reveal how the target is moving, and by the time an action executes, the target is no longer where it was seen: a part on a conveyor at 0.1 m/s travels 15 mm during a 150 ms delay, more than a parallel-jaw grasp typically tolerates (Fig. 1).

Recent work equips policies with a memory of past motion and a forecast of the future [11, 27, 39, 57]. Both cost the resource that dynamic manipulation lacks most, time: richer histories and denser forecasts add computation, and the added latency enlarges the very error the forecast is meant to remove. Existing designs therefore trade information against speed. Video-based WAMs forecast whole future frames or latents at uniform fidelity [27, 44, 50], and methods that restrict prediction to dynamic regions or salient tokens still predict in image space [39, 57]. In both, most capacity goes to appearance, and the quantity that decides contact, the target's metric pose when the action executes, stays implicit. Target-centric predictors sit at the other end [11, 15, 28]: they are fast but blind to surroundings that

can block or redirect the motion. None of them provides knowledge of the moving scene that is precise where contact depends on it, coarse elsewhere, and cheap to use at execution time.

Our key observation is that a dynamic policy needs to know very little precisely. Contact depends on the target's pose at the moment the action executes, to within millimeters. The bodies around it matter only through the clearance they leave and the motion they impart to the target, which a coarse estimate captures, and the rest of the background does not matter. Nor does this knowledge have to be computed at test time: a forecast that supervises training shapes what the policy represents and can be dropped at deployment [54]. SE(3)-WAM builds on both ideas (Fig. 2). It sees the current image together with short metric SE(3) histories of the target and the robot, which convey motion directly instead of a stack of past frames. A world stream is trained to forecast, in one metric 4D space, the target's future SE(3) trajectory precisely and the whole scene coarsely as a sparse 3D flow, and an action stream reads the world stream's summary and replans as often as its latency permits. SE(3)-WAM thus gains the precision of a metric forecast at a latency close to that of a reactive policy.

We evaluate SE(3)-WAM on Dynamic-14, a simulated benchmark of fourteen tasks in which observation, scene motion, and action execution share one physical clock, so the scene keeps moving while a policy computes. SE(3)-WAM achieves state-of-the-art success of [XX.X]%, [XX.X] points above the strongest baseline, at a p95 latency of [XX] ms, and succeeds in [XX]% of trials on six real-robot tasks, against [XX]% for the strongest baseline. Four final-scale controls isolate the contributions of target-motion history, future supervision, and its two geometric components.

Our contributions are threefold:

- We propose role-aware 4D modeling for dynamic manipulation. Tracing how latency and estimation errors reach the contact shows that only the target needs millimeter precision, so we represent the target precisely by its metric SE(3) pose history and future trajectory, and the whole scene coarsely by a sparse 3D flow.
- We present SE(3)-WAM, a world–action model trained with these role-aware 4D forecasts as supervision, at a latency close to that of a reactive policy.
- We build Dynamic-14, in which the scene keeps moving during inference, and show state-of-the-art results on it and on six real-robot tasks.

2. Related Work

Dynamic manipulation. Compensating for the time between sensing and acting is a classical problem, from control with dead time [36] to catching systems that track the target with Kalman-type filters and plan an interception [1–3, 30]; such systems are precise but engineered for each object and task. Learned policies have recently been brought to moving scenes. PUMA reads historical optical flow and is supervised with future object features [11], DynamicWAM conditions a WAM on flow and kinematic descriptors and executes it with real-time chunking [27], and other methods forecast salient tokens, future images, or compact future codes and feed them to the policy at test time [28, 39, 55, 58]. Execution schemes instead reduce or hide latency, by overlapping inference with execution [32, 34, 48], keeping consecutive chunks consistent [6, 25], or conditioning on the execution time [40, 43]. They decide when actions are computed and applied, whereas SE(3)-WAM changes what the policy represents; we run every method under the same real-time executor.

Predictive supervision in world–action models. World–action models (WAMs) generate future video and actions in one network [23, 44, 50, 63], building on video prediction as a policy objective [16, 41, 47]. Because video is costly and dominated by appearance, recent work changes what is forecast, such as dynamic regions, depth, and semantic features [57], flow and geometry alongside RGB [53], object masks or slots [26, 51], or compact latents [21, 22], and asks whether the forecast is needed at test time: Fast-WAM retains most of the benefit of video modeling without test-time imagination [54], and PUMA and others drop their predictors after training [11, 38, 59]. SE(3)-WAM adopts this training-only use and changes the target of the supervision: instead of pixels, features, or latents at one resolution, a metric 4D state whose precision follows each entity's role in contact.

Geometric representations of motion. Metric 3D inputs make policies more precise and sample-efficient [12, 18, 33, 56]. Motion has been represented by 2D point tracks [4, 45, 49, 61], 3D flow [13, 17, 52, 62], and object keypoints or SE(3) trajectories [15, 29, 37, 46], and rigid-body decompositions have long been used to learn scene dynamics [7, 8]. These works describe the scene with one kind of representation, dense flow or tracks everywhere or a pose for each object; DOCIR separates the robot, the target, and obstacles, but as input encodings [10]. SE(3)-WAM assigns precision by role: a coarse sparse 3D flow describes the whole scene, and the target, on which contact depends to within millimeters, is additionally described by its SE(3) history and forecast. Appendix F discusses further related work.

3. SE(3)-WAM

SE(3)-WAM is a world–action model that represents a dynamic scene at role-dependent precision (Fig. 2). It encodes the current RGB-D frame, whose RGB a frozen MAE ViT-

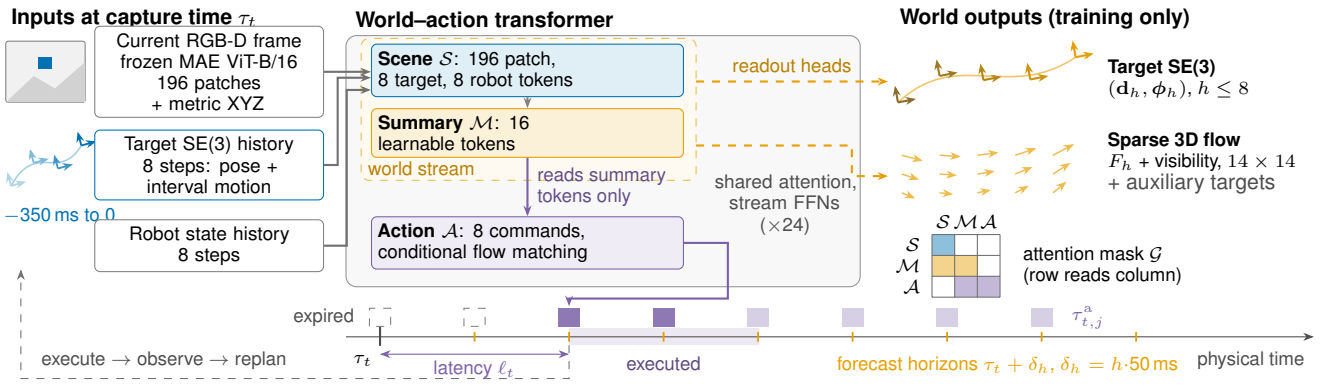


Figure 2. **SE(3)-WAM.** The world stream holds 212 scene tokens, the current image patches from a frozen encoder together with short histories of the target's metric SE(3) pose and of the robot state, and 16 summary tokens. During training, light heads read the world stream out into a role-aware 4D forecast at physical horizons $\tau_t + \delta_h$: the target's SE(3) trajectory and a sparse 3D flow with visibility on a 14×14 grid over the whole scene (dashed: training only). The action stream generates a chunk of 8 commands by flow matching. The attention mask $\mathcal{G}$ lets scene tokens read the scene, summary tokens read the scene and summaries, and actions read summaries and actions. The policy observes and replans at the first control boundary at or after each query ends, where the new chunk takes over from the command meant for that boundary.

B/16 [14] turns into 196 patch tokens and whose depth gives each patch the metric 3D position of its center point in the robot base frame, together with short histories of the target's SE(3) pose and of the robot state. These 212 scene tokens and 16 learnable summary tokens that aggregate them form a world stream, which forecast heads supervise during training with the target's future SE(3) trajectory and a sparse 3D flow of the whole scene. An action stream shares attention with the world stream, reads its summary tokens, and generates action chunks by conditional flow matching [24]. The model receives no past images, so the cost of visual encoding does not grow with the history.

3.1. Problem Formulation

SE(3)-WAM learns a policy from demonstrations. At query t, the policy receives an RGB image, a depth map, and the robot state captured at time τ_t, together with short histories of the robot state and of the pose $T(\tau) = (R(\tau), \mathbf{p}(\tau)) \in \mathrm{SE}(3)$ of a target object in the robot base frame. It outputs a chunk $(\mathbf{a}_{t,0}, \ldots, \mathbf{a}_{t,H-1})$ of commands, one per control period Δ_a, indexed by time from capture: the j-th command is meant for

$$\tau^{\mathrm{a}}_{t,j} = \tau_t + j\Delta_a. \tag{1}$$

The chunk becomes available only after a latency ℓ_t, so its commands with $j\Delta_a < \ell_t$ have expired, and the later ones execute at their times $\tau^{\mathrm{a}}_{t,j}$. The bodies of a scene play different roles. The *target* is the body whose pose must be matched within a contact or containment tolerance ε, such as an object to grasp or a receptacle to place into. The *context* consists of the bodies that constrain or drive the target's motion: obstacles, such as an occluder or a wire, must be avoided with a clearance m, and drivers, such as a conveyor, a ramp, or a striker, move the target. The rest of the scene does not affect the task. A command must therefore fit the target's pose at $\tau^{\mathrm{a}}_{t,j}$ rather than at τ_t, and the lead $\Delta = \tau^{\mathrm{a}}_{t,j} - \tau_t$ of every executed command is at least ℓ_t, so it grows with every millisecond of inference. Sec. 3.2 traces how the error of this estimate depends on the lead and derives the representation of SE(3)-WAM from it.

3.2. Role-Aware 4D Representation

What must be precise. Let a policy act, explicitly or implicitly, on an estimate of the target pose at τ^{a} with position error e and rotation error θ. A contact point at lever arm r is then off by at most $e + 2r\sin(\theta/2)$. Writing the target's motion over the lead as a constant velocity $\mathbf{v}$ plus a residual (acceleration, rolling, impacts), and the lead that the policy assumes as $\hat{\Delta}$, gives

$$e \le e_{\mathrm{loc}} + \hat{\Delta}\, e_v + \|\mathbf{v}\|\, e_\tau + e_{\mathrm{dyn}}, \tag{2}$$

with the errors e_{loc} of the current position, e_v of the velocity, $e_\tau = |\hat{\Delta} - \Delta|$ of the timing, and e_{dyn} of the residual motion (Appendix E). The tolerance ε is typically a few millimeters, and the rotation error is amplified by the lever arm, so the target's pose and motion must be known precisely. The context needs far less precision. An obstacle only has to be avoided with the clearance m, which is usually far larger than ε, and a driver matters through the motion it imparts to the target, which then appears in the target's own future; in both cases, the policy needs to know only roughly where the context moves (Fig. 3c). We therefore represent the target by metric SE(3) states, both its observed history and its forecast trajectory, and the whole scene by a sparse 3D flow, all in the robot base frame and in physical time.

Figure placeholder: Fig. 3 — acting on a moving target (schematic)

Layout: one row, three panels (a)–(c) of equal width, clean vector schematic in the paper's semantic colors; the same robot arm and T-shaped part in all panels; no photographs. **(a) Same pose, different motion:** two histories of the part, A (solid blue) and B (dashed blue), with ghost poses at $-100, -50, 0$ ms that end in the same pose at capture τ_t. Label: "current pose alone is ambiguous". **(b) Pose at execution:** a timeline strip under the panel: capture τ_t, then ticks every $\Delta_a = 50$ ms at the times $\tau^{\mathrm{a}}_{t,j}$ of the chunk's commands; the ticks before the chunk arrives at $\tau_t + \ell_t$ are grayed out as expired, and the first executed command is marked τ^{a}. Above it, the part at τ^{a} for A and B with different positions and rotations (orange triads). Label: "the command must fit the pose at τ^{a}". **(c) Roles:** one scene with three kinds of bodies. *Target*: the part with the end-effector frame E, the target frame O, the contact point, and tolerance ε (mm). *Context, obstacle*: a receptacle wall or occluder with a clearance arrow $m \gg \varepsilon$. *Context, driver*: the conveyor carrying the part, with a motion arrow, and optionally a striker approaching it; label "moves the target". *Background*: table edge and distant objects desaturated in gray; label "no effect". A three-row legend: target → precise SE(3); context → coarse 3D flow; background → nothing. **Rules:** conceptual illustration with hypothetical positions; blue past, orange future, purple command, gray background.

Figure 3. **Acting on a moving target** (schematic). (a) Two histories reach the same pose at capture time τ_t but imply different motions. (b) At the execution time τ^{a}, they lead to different target poses. (c) A contact requires the end-effector frame E to match the target O within ε; an obstacle in the context matters only through the gap $m \gg \varepsilon$ it leaves, a driver such as the conveyor matters through the motion it imparts to the target, and the background does not affect the task.

What latency allows. Latency enters Eq. (2) twice, through the lead that multiplies every velocity error and through the motion that the residual accumulates. Information that costs inference time therefore pays off only if it removes more error than the added latency creates. SE(3)-WAM keeps its inputs cheap, with one current image and low-dimensional state histories instead of past frames, and uses its forecasts only to supervise training, so that they add no inference time (Sec. 3.4).

Target motion: history and forecast. A single pose does not reveal how the target moves (Fig. 3a). The target history covers the last 8 control steps (350 ms at 20 Hz). It consists of the current target pose in the base frame and the seven interval motions between consecutive samples, each with its translation, relative rotation, time interval, and finite-difference velocity and acceleration in physical units, and forms 8 target-state tokens (Appendix A.1). These tokens state directly the velocity and acceleration on which e_v and part of e_{dyn} depend, at the cost of a few low-dimensional tokens. The robot history adds 8 end-effector state tokens. In simulation, the target pose comes from the simulator; on the real robot, a causal tracker estimates it from the RGB-D stream (Appendix C).

The forecast describes where the target will be when the commands execute (Fig. 3b). It covers fixed physical times $\tau_t + \delta_h$, with $\delta_h = h \cdot 50$ ms and $h = 1, \dots, 8$, from 50 ms to 400 ms after capture. Like all our motion labels, it is cumulative from the current state, and every loss averages over valid labels only. For each horizon, the target's motion is

$$\mathbf{d}_h = \mathbf{p}(\tau_t+\delta_h) - \mathbf{p}(\tau_t), \quad \boldsymbol{\phi}_h = \mathrm{Log}\big(R(\tau_t+\delta_h)R(\tau_t)^\top\big)^\vee. \tag{3}$$

A linear pose head on the mean of the final summary tokens predicts $(\hat{\mathbf{d}}_h, \hat{\boldsymbol{\phi}}_h)$ with the loss

$$\mathcal{L}_P = \frac{1}{|\mathcal{V}|} \sum_{h \in \mathcal{V}} \Big[\|\hat{\mathbf{d}}_h - \mathbf{d}_h\|^2 + \lambda_R \, d_{\mathrm{SO}(3)}\big(\hat{R}_h, R_h\big)^2 \Big], \tag{4}$$

where $\hat{R}_h = \mathrm{Exp}(\hat{\boldsymbol{\phi}}_h^\wedge) R(\tau_t)$ and $\mathcal{V}$ is the set of valid horizons. Because the pose head reads the summary tokens, the only world tokens that the actions attend to (Sec. 3.3), the precise target supervision shapes exactly the representation that the actions use. The past and the future of the target thus share one metric frame and one time axis.

Scene motion: sparse 3D flow. The center points of the 196 patches on the 14×14 grid cover the target, the robot, and the surroundings. For each point i and horizon h, a linear head on its final patch token predicts the base-frame displacement $\hat{F}_{ih} \in \mathbb{R}^3$ and a visibility logit $\hat{o}_{ih}$, and

$$\mathcal{L}_F = \frac{1}{|\mathcal{V}_F|} \sum_{(i,h) \in \mathcal{V}_F} \Big[\tfrac{1}{3} \big\| (\hat{F}_{ih} - F_{ih})/s \big\|^2 + \mathrm{BCE}(\hat{o}_{ih}, o_{ih}) \Big], \tag{5}$$

with a fixed scale s fitted on the training labels, the visibility o_{ih}, and the set $\mathcal{V}_F$ of valid labels. The grid samples the workspace sparsely, about [X] mm between points. The flow thus gives the context and the robot a motion estimate at this spacing and stays near zero on the static background, while the target additionally receives the dedicated SE(3) supervision of Eq. (4). The flow supervises the patch tokens from which the summaries draw, one level below the precise target supervision; this is how SE(3)-WAM allocates precision by role.

Auxiliary supervision. Two further targets regularize the representation; we selected them on development scenes (Appendix D.5). Future appearance heads on the patch tokens predict, at every horizon, the metric depth, the features of a frozen DINOv3 teacher [35], and the RGB of moving surfaces. Separately, a small head on the summary tokens reconstructs the RGB of the current target region, which ties the summaries to the target without a future label. Their losses form $\mathcal{L}_{\mathrm{aux}}$ (Appendix A.2).

3.3. Action Generation and Joint Learning

Action stream. The action stream holds $H = 8$ tokens, one per command of the chunk; each embeds a noisy command, an end-effector translation and rotation increment plus

a gripper command (Appendix A.3). The attention mask $\mathcal{G}$ lets scene tokens attend to the scene, summary tokens to the scene and the summaries, and action tokens to the summaries and the actions, so the actions read the world through the summary tokens and the world stream does not depend on the actions. In every layer, both streams share the attention projections, through which they exchange information, and keep separate feed-forward networks, which give each its own capacity; we selected this backbone over fully separate parameters on development scenes (Appendix D.5). World tokens use layer normalization, and action tokens use adaptive layer normalization conditioned on the flow time.

Joint learning. For a chunk $\mathbf{a}_1$ normalized with training statistics (Appendix A.3), noise $\mathbf{a}_0 \sim \mathcal{N}(0, I)$, and flow time $u \sim \mathcal{U}[0, 1]$, the action stream v_ψ, with parameters ψ, regresses the flow-matching velocity

$$\mathcal{L}_A = \mathbb{E} \left\| v_\psi(\mathbf{a}_u, u \mid \mathcal{S}_t, \mathcal{G}) - (\mathbf{a}_1 - \mathbf{a}_0) \right\|^2, \qquad (6)$$

with $\mathbf{a}_u = (1-u)\mathbf{a}_0 + u\mathbf{a}_1$, the scene tokens $\mathcal{S}_t$ of query t, and the error averaged over valid commands. All parts are trained jointly with

$$\mathcal{L} = \mathcal{L}_A + \lambda_P \mathcal{L}_P + \lambda_F \mathcal{L}_F + \lambda_{\text{aux}} \mathcal{L}_{\text{aux}}, \qquad (7)$$

with the weights given in Sec. 4.1.

3.4. Efficient Inference and Execution

SE(3)-WAM trains its forecast heads jointly and skips them at test time, as recent WAMs do [23, 54]. Because world tokens attend neither to the actions nor to the flow time, the world stream runs once per query, and the action stream integrates 5 Euler steps on its cached keys and values. The executor counts latency in whole control periods, as RTC does [6]. A query of total latency L_t hands its chunk over at the first control-period boundary at or after it ends, $\ell_t = k_t \Delta_a$ after capture with $k_t = \lceil L_t / \Delta_a \rceil$, and execution continues from command k_t, the one meant for that boundary. There the policy observes again and starts the next query; until then, the executor continues the previous chunk. The policy therefore replans as often as its latency allows, before every command when a query fits in one control period. Because each command executes at the time its label refers to, the action stream needs no latency input, and every executed command, at most $(H-1)\Delta_a$ after capture, lies within the horizons of the forecast supervision (Appendix B.3). On the robot, L_t also includes tracking the target in the newest frame.

4. Experiments

We compare SE(3)-WAM with reactive, dynamic, and world–action policies in simulation (Sec. 4.2) and on a real robot (Sec. 4.3), and then ablate its target history and its geometric supervision (Sec. 4.4).

4.1. Setup

Dynamic-14. We built fourteen tasks in MuJoCo/robosuite [42, 64] in five families: grasping a moving target (harmonic, conveyor, translating, rotating, and occluded motion), placing into a moving receptacle, passing a wand along a rotating wire, capturing rolling or sliding objects, and grasping after an impact (Appendix B, Fig. S2). Experts with privileged simulator state provide 800 demonstrations per task. We select configurations on development scenes and report all Dynamic-14 results on 1,400 held-out scenes (100 per task, new seeds). Our variants average three training seeds, and each baseline is trained once. We give paired 95% intervals from a bootstrap over held-out scenes.

Protocol. Observation, scene motion, and execution share one physical clock with a control period of 50 ms, and physics advances during inference by a latency drawn from each method's measured latency on one RTX 5090. All methods run under the same executor: a query starts at a control-period boundary, its chunk takes over at the first boundary at or after the query ends, from the command meant for that boundary, and the next query starts there (Appendix B.3). The +150 ms setting adds 150 ms of delay for every method, and the paused setting stops physics during inference. We report the task-balanced success rate (SR), the position error at first contact e_p, and the p95 time-to-action.

Baselines. We compare with reactive policies (ACT [60], Diffusion Policy [9], and $\pi_{0.5}$ [31]), dynamic VLAs (DynamicVLA [48] and PUMA [11]), image- or latent-space WAMs (DreamVLA [57] and OpenWAM-α [44]), and a dynamic WAM (DynamicWAM [27]). Each baseline uses its official code, inputs, and recipe, is trained on the same demonstrations, and receives a rendered wrist camera if its recipe uses one (Appendix B.4). SE(3)-WAM uses only the agent view, together with depth and the target pose. To give baselines the same target-motion information, $\pi_{0.5}$ and DynamicWAM also run with our SE(3) history tokens (+SE(3)). The *CV input* variant is SE(3)-WAM without forecast losses that instead receives the constant-velocity extrapolation of the target pose as input tokens.

Implementation. The transformer has 24 layers of width 1024 with 16 heads and 523.8M trainable parameters; the MAE ViT-B/16 encoder and the DINOv3 teacher stay frozen. On Dynamic-14, we train from a randomly initialized transformer on all windows of the demonstrations for 20 epochs (51,740 updates at a global batch size of 512) with AdamW, a cosine schedule, $\lambda_P = \lambda_F = 1$, and $\lambda_R = \lambda_{\text{aux}} = 0.1$, which takes [X] hours per seed on [N] RTX PRO 6000 GPUs (Appendix A.4). For the real robot, we pretrain a 48-layer base (1.02B parameters) on DROID [19], whose demonstra-

Figure placeholder: Fig. 4 — real-robot trials

Rows: a success of SE(3)-WAM; the strongest baseline on the same trial key; a failure of SE(3)-WAM. **Columns:** the query frame, the first execution time τ^a, and contact. **Overlays:** blue tracked past poses; purple commanded gripper path. Real photos, unretouched; overlays from logged data of each trial.

Figure 4. **Real-robot trials** of SE(3)-WAM and [the strongest baseline] on the same trial key (top two rows) and a failure of SE(3)-WAM (bottom). [One-sentence description of the selected trials.]

tions carry no target poses: it learns the actions, the sparse 3D flow from the point tracks that PointWorld released [17], and the future DINOv3 features, for 21,060 updates on 64 GPUs. Fine-tuning on each task adds the target-state tokens and the pose head (Appendix A.5).

4.2. Simulation Results

SE(3)-WAM reaches [XX.X]% SR on held-out Dynamic-14 scenes (Table 1), [X.X] points above the strongest baseline, [name of the strongest baseline], and [X.X] points above the strongest baseline that receives the same SE(3) history, [name]. The margin is largest on [rotating targets and on fast conveyor and translating targets], where the target moves farthest during the lead, and its position error at first contact is [X.X] mm against [X.X] mm for [name]. Its p95 time-to-action of [XX] ms is [X]× lower than that of the image-space WAMs, whose latency adds to the lead they must bridge. With 150 ms of added delay, SE(3)-WAM keeps [XX.X]%, whereas reactive policies drop to [XX.X]% and image-space WAMs to [XX.X]%. With physics paused during inference, the difference to [name of the strongest paused baseline] is [±X.X] points, which separates the contribution of speed from that of the representation. The CV-input variant, which extrapolates the target's current motion instead of learning a forecast, trails SE(3)-WAM by [X.X] points.

DOMINO. On a second benchmark, we compare SE(3)-WAM with $\pi_{0.5}$, DynamicWAM, and PUMA on 24 single-arm Franka tasks of DOMINO [11], with its demonstrations and native evaluator. SE(3)-WAM reaches [XX.X]% SR against [XX.X]% for [the strongest baseline] (Appendix D.3). As on Dynamic-14, it reads the simulator pose of the target.

4.3. Real-Robot Experiments

We evaluate six tasks on a Franka arm ([model]) with [gripper model] and a fixed RealSense D435 RGB-D camera (Fig. S3, Table S8), comparing SE(3)-WAM with $\pi_{0.5}$, DynamicWAM, and Diffusion Policy [9]: conveyor grasping (T1), placement into a receptacle on the conveyor (T2), turntable grasping (T3), grasping a moving toy car (T4), intercepting a ball after a collision (T5), and intercepting an object that rolls or slides down a ramp (T6). Together they cover linear, circular, free planar, post-impact, and gravity-accelerated target motion. Each method is trained or fine-tuned per task on the same 100 teleoperated demonstrations, with 20 separate validation demonstrations, and runs 20 trials per task with shared trial keys. SE(3)-WAM tracks the target with [tracker] and runs on one RTX 5090 with a p95 time-to-action of [XX] ms. It succeeds in [XX.X]% of all trials, against [XX.X]% for [the strongest baseline] (Table 2). [Where its advantage is largest, and its dominant failure mode] (Fig. 4).

4.4. Ablations

Four controls each change one component of the final model and keep the data, recipe, and training budget, with three training seeds each (Table 3, Fig. 5). We evaluate them under the real-time protocol at 0 and 150 ms of added delay.

Target history. The no-history control keeps the current image, the current target pose, and the robot history, and removes the past target poses with their velocity and acceleration features. Without them, SR changes by [signed X.X] points at 0 ms and [signed X.X] points at 150 ms of added delay (Fig. 5a). Under synthetic pose noise, SE(3)-WAM keeps [XX.X]% SR (Appendix D.4).

Future supervision. The no-future control removes all forecast losses, the SE(3), scene-flow, and future appearance targets, and keeps the current-target RGB reconstruction. It changes SR by [signed X.X] points at 0 ms and [signed X.X] points at 150 ms, which measures the effect of future supervision as a whole.

The two geometric components. The other two controls keep all auxiliary targets. Removing the scene-flow supervision changes SR by [signed X.X] points; since the flow covers the whole scene, this measures coarse scene motion as a whole rather than the surroundings alone. Removing the dedicated SE(3) supervision leaves the target with only the coarse flow that every part of the scene receives and changes SR by [signed X.X] points, most on [task families]. Fig. 6 shows what the SE(3) head predicts at the execution time.

5. Conclusion

We presented SE(3)-WAM, a world–action model for manipulating moving objects that spends precision where contact depends on it and time only where it pays off. It remembers the target's metric SE(3) motion instead of past frames, learns to forecast the target's SE(3) trajectory precisely and

Table 1. **Main results.** Held-out SR (%) on Dynamic-14 (1,400 new scenes, task-balanced, real time, three training seeds for our variants and one for baselines) by family, with 150 ms of added delay, and with physics paused during inference, together with the position error at first contact e_p (mm) and the p95 time-to-action on one RTX 5090. Inputs: R, RGB and robot state. D, depth. P, target SE(3) history. Best in bold. Intervals in Appendix D.

		Dynamic-14 SR, real time unless noted									
Method	Inputs	All	Grasp	Place	Pass	Capture	Impact	+150 ms	Paused	e_p	p95 (ms)
ACT [60]	R	—	—	—	—	—	—	—	—	—	—
Diffusion Policy [9]	R	—	—	—	—	—	—	—	—	—	—
$\pi_{0.5}$ [31]	R	—	—	—	—	—	—	—	—	—	—
$\pi_{0.5}$ + SE(3)	R+P	—	—	—	—	—	—	—	—	—	—
DynamicVLA [48]	R	—	—	—	—	—	—	—	—	—	—
PUMA [11]	R	—	—	—	—	—	—	—	—	—	—
DreamVLA [57]	R	—	—	—	—	—	—	—	—	—	—
OpenWAM-α [44]	R	—	—	—	—	—	—	—	—	—	—
DynamicWAM [27]	R	—	—	—	—	—	—	—	—	—	—
DynamicWAM + SE(3)	R+P	—	—	—	—	—	—	—	—	—	—
SE(3)-WAM, CV input	RD+P	—	—	—	—	—	—	—	—	—	—
SE(3)-WAM	RD+P	—	—	—	—	—	—	—	—	—	—

Figure placeholder: Fig. 5 — ablations (two panels)

(a) By task family. SR difference to the full model of each of the four controls, per task family, at 0 and 150 ms added delay. **(b) Success and latency.** p95 time-to-action vs. SR for the full model and the four controls, at 0 and 150 ms added delay. All panels reuse the same held-out scenes and three training seeds; show paired 95% bootstrap intervals.

Figure 5. **Ablations.** (a) Effect of each control per task family. (b) Success and latency of the full model and the controls under the real-time protocol. [Summarize the measured effects.]

Table 2. **Real-robot results.** Success rate (%) on six tasks: T1 conveyor grasping, T2 moving-receptacle placement, T3 turntable grasping, T4 moving-car grasping, T5 post-collision interception, and T6 ramp interception (20 trials per task and method). All pools the trials; intervals in Appendix C.

Method	T1	T2	T3	T4	T5	T6	All
$\pi_{0.5}$	—	—	—	—	—	—	—
DynamicWAM	—	—	—	—	—	—	—
Diffusion Policy	—	—	—	—	—	—	—
SE(3)-WAM	—	—	—	—	—	—	—

Figure placeholder: Fig. 6 — forecasts at the execution time

Rows: a rolling capture (median forecast error) and a post-impact grasp. **Columns:** the query frame at τ_t, the frame at τ^a, and a 3D inset in the base frame. **Overlays:** blue past poses, orange forecast triads and coarse flow arrows from the training-time heads, gray ground truth at the same times, purple commanded path. Full set in Fig. S5.

Figure 6. **Forecasts at the execution time** on held-out Dynamic-14 episodes, read from the training-time heads. Top: the SE(3) forecast follows a rolling target that constant-velocity extrapolation overshoots. Bottom: before an impact, the regressed forecast lies between the possible outcomes.

the whole scene coarsely as a sparse 3D flow, and lets its actions read a representation shaped by these forecasts without generating them at test time. On Dynamic-14, where the scene keeps moving while a policy computes, and on six real-robot tasks, it improves on reactive, dynamic, and world–action policies. Four final-scale controls examine the metric motion history, future supervision, and its two geometric components. [State the supported component-level

Table 3. **Core ablations** of the final 24-layer model on held-out Dynamic-14 scenes (three training seeds). Each row changes one component of SE(3)-WAM (*). SR, SR_{150}: success (%) without and with 150 ms of added delay. e_p: position error at first contact (mm). p95: time-to-action (ms).

Variant	SR	SR_{150}	e_p	p95
SE(3)-WAM *	—	—	—	—
No target history	—	—	—	—
No future supervision	—	—	—	—
No scene-flow supervision	—	—	—	—
No dedicated SE(3)	—	—	—	—

conclusions after the core ablations.]

Limitations. SE(3)-WAM assumes a rigid target that moves on its own until contact, so deformable or articulated targets and post-contact dynamics are outside its scope. It depends on a pose estimate of the target, and tracker failures propagate to the policy; in simulation we read the pose from the simulator. The regressed target forecast cannot represent multimodal futures, e.g., before an impact. Our real-robot base learns from DROID without target poses, so its pose-related parts are learned only during fine-tuning. Finally, our tasks have a single target and slow context motion.

LLM Usage Statement

[To be completed before submission: actual tools used, affected parts of the research and manuscript, and extent of use.]

References

[1] Peter K. Allen, Aleksandar Timcenko, Billibon Yoshimi, and Paul Michelman. Automated tracking and grasping of a moving object with a robotic hand-eye system. *IEEE Transactions on Robotics and Automation*, 9(2):152–165, 1993. 2

[2] Yaakov Bar-Shalom, X. Rong Li, and Thiagalingam Kirubarajan. *Estimation with Applications to Tracking and Navigation*. Wiley, 2001.

[3] Berthold Bäuml, Thomas Wimböck, and Gerd Hirzinger. Kinematically optimal catching a flying ball with a hand-arm-system. In *IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2010. 2

[4] Homanga Bharadhwaj, Roozbeh Mottaghi, Abhinav Gupta, and Shubham Tulsiani. Track2Act: Predicting point tracks from internet videos enables generalizable robot manipulation. In *European Conference on Computer Vision*, 2024. 2

[5] Kevin Black, Noah Brown, Danny Driess, et al. π_0: A vision-language-action flow model for general robot control. In *Proceedings of Robotics: Science and Systems*, 2025. 1

[6] Kevin Black, Manuel Y. Galliker, and Sergey Levine. Real-time execution of action chunking flow policies. In *Advances in Neural Information Processing Systems*, 2025. 2, 5

[7] Arunkumar Byravan and Dieter Fox. SE3-Nets: Learning rigid body motion using deep neural networks. In *IEEE International Conference on Robotics and Automation*, 2017. 2

[8] Arunkumar Byravan, Felix Leeb, Franziska Meier, and Dieter Fox. SE3-Pose-Nets: Structured deep dynamics models for visuomotor control. In *IEEE International Conference on Robotics and Automation*, 2018. 2

[9] Cheng Chi, Siyuan Feng, Yilun Du, Zhenjia Xu, Eric Cousineau, Benjamin C. M. Burchfiel, and Shuran Song. Diffusion policy: Visuomotor policy learning via action diffusion. In *Proceedings of Robotics: Science and Systems*, 2023. 1, 5, 6, 7

[10] David Emukpere, Romain Deffayet, Bingbing Wu, Romain Brégier, Michael Niemaz, Jean-Luc Meunier, Denys Proux, Jean-Michel Renders, and Seungsu Kim. Disentangled object-centric image representation for robotic manipulation. *arXiv preprint arXiv:2503.11565*, 2025. 2

[11] Heng Fang, Shangru Li, Shuhan Wang, Xuanyang Xi, Dingkang Liang, and Xiang Bai. Towards generalizable robotic manipulation in dynamic environments. *arXiv preprint arXiv:2603.15620*, 2026. 1, 2, 5, 6, 7

[12] Ankit Goyal, Jie Xu, Yijie Guo, Valts Blukis, Yu-Wei Chao, and Dieter Fox. RVT: Robotic view transformer for 3D object manipulation. In *Conference on Robot Learning*, pages 694–710, 2023. 2

[13] Jun Guo, Xiaojian Ma, Yikai Wang, Min Yang, Huaping Liu, and Qing Li. FlowDreamer: A RGB-D world model with flow-based motion representations for robot manipulation. *arXiv preprint arXiv:2505.10075*, 2025. 2

[14] Kaiming He, Xinlei Chen, Saining Xie, Yanghao Li, Piotr Dollár, and Ross Girshick. Masked autoencoders are scalable vision learners. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 16000–16009, 2022. 3

[15] Cheng-Chun Hsu, Bowen Wen, Jie Xu, Yashraj Narang, Xiaolong Wang, Yuke Zhu, Joydeep Biswas, and Stan Birchfield. SPOT: SE(3) pose trajectory diffusion for object-centric manipulation. In *IEEE International Conference on Robotics and Automation*, 2025. 1, 2

[16] Yucheng Hu, Yanjiang Guo, Pengchao Wang, Xiaoyu Chen, Yen-Jen Wang, Jianke Zhang, Koushil Sreenath, Chaochao Lu, and Jianyu Chen. Video prediction policy: A generalist robot policy with predictive visual representations. In *International Conference on Machine Learning*, 2025. 2

[17] Wenlong Huang, Yu-Wei Chao, Arsalan Mousavian, Ming-Yu Liu, Dieter Fox, Kaichun Mo, and Li Fei-Fei. PointWorld: Scaling 3D world models for in-the-wild robotic manipulation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2026. 2, 6

[18] Tsung-Wei Ke, Nikolaos Gkanatsios, and Katerina Fragkiadaki. 3D diffuser actor: Policy diffusion with 3D scene representations. In *Conference on Robot Learning*, 2024. 2

[19] Alexander Khazatsky, Karl Pertsch, Suraj Nair, et al. DROID: A large-scale in-the-wild robot manipulation dataset. In *Proceedings of Robotics: Science and Systems*, 2024. 5

[20] Moo Jin Kim, Karl Pertsch, Siddharth Karamcheti, et al. OpenVLA: An open-source vision-language-action model. In *Conference on Robot Learning*, 2024. 1

[21] Fenghao Lei, Zhixiong Huang, Long Yang, Jiabao Chen, Peilin Huang, Han Fu, Zhuo Li, and Xiaoxue Ren. DELE-w0.5: Inferring action from future latent state for robotic manipulation. *arXiv preprint arXiv:2608.22067*, 2026. 2

[22] Jiajun Li, Tiecheng Guo, Yifan Ye, Rongyu Zhang, Xiaowei Chi, Qianpu Sun, Ying Li, Yunfan Lou, Jinqiao Wang, Zhihe Lu, Meng Guo, and Shanghang Zhang. Efficient-WAM: A 1B-parameter world-action model with low-cost future imagination. *arXiv preprint arXiv:2606.10040*, 2026. 2

[23] Shuang Li, Yihuai Gao, Dorsa Sadigh, and Shuran Song. Unified video action model. In *Proceedings of Robotics: Science and Systems*, 2025. 2, 5

[24] Yaron Lipman, Ricky T. Q. Chen, Heli Ben-Hamu, Maximilian Nickel, and Matthew Le. Flow matching for generative modeling. In *International Conference on Learning Representations*, 2023. 3

[25] Yuejiang Liu, Jubayer Ibn Hamid, Annie Xie, Yoonho Lee, Maximilian Du, and Chelsea Finn. Bidirectional decoding: Improving action chunking via guided test-time sampling. In *International Conference on Learning Representations*, 2025. 2

[26] Yushan Liu, Peibo Sun, Shoujie Li, Yifan Xie, Lingfeng Zhang, Xintao Chao, Shiyuan Dong, Fang Chen, Xiao-Ping Zhang, and Wenbo Ding. OA-WAM: Object-addressable world action model for robust robot manipulation. *arXiv preprint arXiv:2605.06481*, 2026. 2

[27] Yunfan Lou, Hewen Gao, Xiyu Zhu, Zhuoran Qiao, Xuan Han, Yifan Yang, Yifan Ye, Boxian Yao, and Zhibo Pang. DynamicWAM: Dual-path motion conditioning for world-action models in dynamic manipulation. *arXiv preprint arXiv:2608.00793*, 2026. 1, 2, 5, 7

[28] Siyuan Ma, Yutian Zhang, Boshi Zhang, Qinglian Wu, Jiaqi Zhai, Dong Wei, and Xiaojin Huang. ForeTime-VLA: Causal future-token distillation from a world action model for conveyor-belt manipulation. *arXiv preprint arXiv:2608.20735*, 2026. 1, 2

[29] Lucas Manuelli, Wei Gao, Peter Florence, and Russ Tedrake. kPAM: Keypoint affordances for category-level robotic manipulation. In *International Symposium on Robotics Research*, 2019. 2

[30] Naresh Marturi, Marek Kopicki, Alireza Rastegarpanah, et al. Dynamic grasp and trajectory planning for moving objects. *Autonomous Robots*, 43(5):1241–1256, 2019. 2

[31] Physical Intelligence, Kevin Black, Noah Brown, James Darpinian, et al. $\pi_{0.5}$: a vision-language-action model with open-world generalization. *arXiv preprint arXiv:2504.16054*, 2025. 1, 5, 7

[32] Yiran Shi, Dongqi Guo, Tianchen Zhao, Feng Gao, Liangzhi Shi, Chao Yu, ZhiJian Mo, Qihua Xiao, XiaoShuai Peng, Qingmin Liao, and Yu Wang. StreamingVLA: Streaming vision-language-action model with action flow matching and adaptive early observation. *arXiv preprint arXiv:2603.28565*, 2026. 2

[33] Mohit Shridhar, Lucas Manuelli, and Dieter Fox. Perceiver-actor: A multi-task transformer for robotic manipulation. In *Conference on Robot Learning*, 2022. 2

[34] Mustafa Shukor, Dana Aubakirova, Francesco Capuano, Pepijn Kooijmans, Steven Palma, et al. SmolVLA: A vision-language-action model for affordable and efficient robotics. *arXiv preprint arXiv:2506.01844*, 2025. 2

[35] Oriane Siméoni, Huy V. Vo, Maximilian Seitzer, Federico Baldassarre, Maxime Oquab, Cijo Jose, Vasil Khalidov, Marc Szafraniec, Seungeun Yi, Michaël Ramamonjisoa, et al. DINOv3. *arXiv preprint arXiv:2508.10104*, 2025. 4

[36] Otto J. M. Smith. Closer control of loops with dead time. *Chemical Engineering Progress*, 53(5):217–219, 1957. 2

[37] Rustin Soraki, Homanga Bharadhwaj, Ali Farhadi, and Roozbeh Mottaghi. ObjectForesight: Predicting future 3D object trajectories from human videos. *arXiv preprint arXiv:2601.05237*, 2026. 2

[38] Yue Su, Sijin Chen, Haixin Shi, Mingyu Liu, Zhengshen Zhang, Ningyuan Huang, Weiheng Zhong, Zhengbang Zhu, Yuxiao Liu, and Xihui Liu. World guidance: World modeling in condition space for action generation. *arXiv preprint arXiv:2602.22010*, 2026. 2

[39] Shahram Najam Syed, Arthur Jakobsson, Haoran Hao, and Jeffrey Ichnowski. Intercepting the future: Latent-space predictive world model for dynamic VLA manipulation. *arXiv preprint arXiv:2606.02486*, 2026. 1, 2

[40] Jiaming Tang, Yufei Sun, Yilong Zhao, Shang Yang, Yujun Lin, Zhuoyang Zhang, James Hou, Yao Lu, Zhijian Liu, and Song Han. VLASH: Real-time VLAs via future-state-aware asynchronous inference. *arXiv preprint arXiv:2512.01031*, 2025. 2

[41] Yang Tian, Sizhe Yang, Jia Zeng, Ping Wang, Dahua Lin, Hao Dong, and Jiangmiao Pang. Predictive inverse dynamics models are scalable learners for robotic manipulation. In *International Conference on Learning Representations*, 2025. 2

[42] Emanuel Todorov, Tom Erez, and Yuval Tassa. MuJoCo: A physics engine for model-based control. In *IEEE/RSJ International Conference on Intelligent Robots and Systems*, pages 5026–5033, 2012. 5

[43] Changhao Wang, Jaden Clark, and Shuran Song. Chronos policy: Learning robot policies on physical time. *arXiv preprint arXiv:2610.13929*, 2026. 2

[44] Yuran Wang, Siqiao Huang, Mingleyang Li, Chenhao Zhang, Jiaqi Liang, Weiyang Jin, Yue Chen, Xuemin Chi, Donghao Zhou, Qize Yu, Yu-Kai Wang, Yuhan Rui, Shenzhe Yao, Zhen Yuan, Zhenhao Shen, Kefei Zhu, Zijie Zhu, Ning Gao, Xiaowei Chi, Guanqi He, Shanghang Zhang, Hao Dong, Lin Shao, and Hang Zhao. OpenWAM: An open, modular exploration towards systematic world-action model pretraining. *arXiv preprint arXiv:2609.07398*, 2026. 1, 2, 5, 7

[45] Chuan Wen, Xingyu Lin, John So, Kai Chen, Qi Dou, Yang Gao, and Pieter Abbeel. Any-point trajectory modeling for policy learning. In *Proceedings of Robotics: Science and Systems*, 2024. 2

[46] Gaoyuan Wu, Ziyu Shan, Haoyang Du, Yuyao Jiang, and Ziwei Wang. AffordTrajDP: Dynamic affordance-guided visuomotor policy learning for robotic manipulation. *arXiv preprint arXiv:2608.01603*, 2026. 2

[47] Hongtao Wu, Ya Jing, Chilam Cheang, Guangzeng Chen, Jiafeng Xu, Xinghang Li, Minghuan Liu, Hang Li, and Tao Kong. Unleashing large-scale video generative pre-training for visual robot manipulation. In *International Conference on Learning Representations*, 2024. 2

[48] Haozhe Xie, Beichen Wen, Jiarui Zheng, Zhaoxi Chen, Fangzhou Hong, Haiwen Diao, and Ziwei Liu. DynamicVLA: A vision-language-action model for dynamic object manipulation. *arXiv preprint arXiv:2601.22153*, 2026. 2, 5, 7

[49] Mengda Xu, Zhenjia Xu, Yinghao Xu, Cheng Chi, Gordon Wetzstein, Manuela Veloso, and Shuran Song. Flow as the cross-domain manipulation interface. In *Conference on Robot Learning*, 2024. 2

[50] Seonghyeon Ye, Yunhao Ge, Kaiyuan Zheng, Shenyuan Gao, Sihyun Yu, George Kurian, Suneel Indupuru, et al. World action models are zero-shot policies. *arXiv preprint arXiv:2602.15922*, 2026. 1, 2

[51] Hanyang Yu, Haitao Lin, Jingbo Zhang, Wenyao Zhang, Chenghao Gu, Heng Li, and Ping Tan. MaskWAM: Unifying mask prompting and prediction for world-action models. *arXiv preprint arXiv:2606.13515*, 2026. 2

[52] Chengbo Yuan, Chuan Wen, Tong Zhang, and Yang Gao. General flow as foundation affordance for scalable robot learning. In *Conference on Robot Learning*, pages 1541–1566, 2024. 2

[53] Shanglin Yuan, Weiheng Zhao, Xin Shi, Haoyi Jiang, Xianda Guo, Liu Liu, Wenyu Liu, Wei Sui, and Xinggang Wang. DreamWAM: Beyond RGB future prediction for world action models. *arXiv preprint arXiv:2608.04996*, 2026. 2

[54] Tianyuan Yuan, Zibin Dong, Yicheng Liu, and Hang Zhao. Fast-WAM: Do world action models need test-time future imagination? *arXiv preprint arXiv:2603.16666*, 2026. 2, 5

[55] Peng Yun, Shouwang Huang, Hao Li, Jinxi Li, Jianan Wang, and Bo Yang. PhysMani: Physics-principled 3D world model for dynamic object manipulation. *arXiv preprint arXiv:2607.01938*, 2026. 2

[56] Yanjie Ze, Gu Zhang, Kangning Zhang, Chenyuan Hu, Muhan Wang, and Huazhe Xu. 3D diffusion policy: Generalizable visuomotor policy learning via simple 3D representations. In *Proceedings of Robotics: Science and Systems*, 2024. 2

[57] Wenyao Zhang, Hongsi Liu, Zekun Qi, Yunnan Wang, Xin-Qiang Yu, Jiazhao Zhang, Runpei Dong, Jiawei He, He Wang, Zhizheng Zhang, Li Yi, Wenjun Zeng, and Xin Jin. DreamVLA: A vision-language-action model dreamed with comprehensive world knowledge. In *Advances in Neural Information Processing Systems*, pages 24195–24228, 2025. 1, 2, 5, 7

[58] Yifan Zhang, Ruiping Wang, and Xilin Chen. Dynamic behavior cloning with temporal feature prediction: Enhancing robotic arm manipulation in moving object tasks. *IEEE Robotics and Automation Letters*, 10(6):5209–5216, 2025. 2

[59] Ruiteng Zhao, Zhengshen Zhang, Yue Su, Wenshuo Wang, Jiahui Li, Zhiyuan Yang, Francis E. H. Tay, Marcelo H. Ang, and Haiyue Zhu. SG-WAM: Self-guided world modeling in geometry-aware policy space. *arXiv preprint arXiv:2608.01397*, 2026. 2

[60] Tony Z. Zhao, Vikash Kumar, Sergey Levine, and Chelsea Finn. Learning fine-grained bimanual manipulation with low-cost hardware. In *Proceedings of Robotics: Science and Systems*, 2023. 5, 7

[61] Ruijie Zheng, Yongyuan Liang, Shuaiyi Huang, Jianfeng Gao, Hal Daumé III, Andrey Kolobov, Furong Huang, and Jianwei Yang. TraceVLA: Visual trace prompting enhances spatial-temporal awareness for generalist robotic policies. In *International Conference on Learning Representations*, 2025. 2

[62] Hongyan Zhi, Peihao Chen, Siyuan Zhou, Yubo Dong, Quanxi Wu, Lei Han, and Mingkui Tan. 3DFlowAction: Learning cross-embodiment manipulation from 3D flow world model. *arXiv preprint arXiv:2506.06199*, 2025. 2

[63] Chuning Zhu, Raymond Yu, Siyuan Feng, Benjamin Burchfiel, Paarth Shah, and Abhishek Gupta. Unified world models: Coupling video and action diffusion for pretraining on large robotic datasets. In *Proceedings of Robotics: Science and Systems*, 2025. 2

[64] Yuke Zhu, Josiah Wong, Ajay Mandlekar, Roberto Martín-Martín, Abhishek Joshi, Kevin Lin, Abhiram Maddukuri, Soroush Nasiriany, and Yifeng Zhu. robosuite: A modular simulation framework and benchmark for robot learning. *arXiv preprint arXiv:2009.12293*, 2020. 5

[65] Brianna Zitkovich, Tianhe Yu, Sichun Xu, et al. RT-2: Vision-language-action models transfer web knowledge to robotic control. In *Conference on Robot Learning*, 2023. 1